Lesson 17 Relations

Example "Equality"

numbers : $x = y$ same value

sets : $A = B$ same elements/members

statements : $p \equiv q$ same T.T./"meaning"

Example : "$\leq$"

numbers : $x \leq y$

sets : $A \subseteq B$

statements : $p \to q$

Def Let A and B be sets.

- R is a relation on A if $R \subseteq A \times A$.

  $A \times A = \{(x,y) : x, y \in A\}$

- R is a relation from A to B if $R \subseteq A \times B$.

$\boxed{(x,y) \in R}$ iff $\boxed{xRy}$: x is related to y

$(x,y) \notin R \leftrightarrow x \not R\, y$: x is not related to y

Ex Let R be "$=$" relation on $\mathbb{Z}$.

$R = \{\ldots\ldots, (-2,-2), (-1,-1), (0,0), (1,1), \ldots\}$

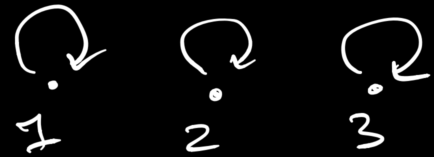
$1 R 1 \leftrightarrow (1,1) \in R \leftrightarrow 1 = 1$

$1 \not R\, 2 \leftrightarrow (1,2) \notin R \leftrightarrow 1 \neq 2$

Picture

$A = \{1, 2, 3\}$, R : "$=$" on A

$R = \{(1,1),(2,2),(3,3)\}$



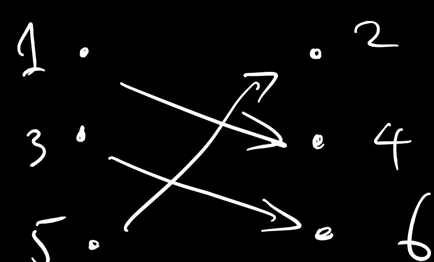
$R = \{(x,y)\in A\times A : x<y\}$

$= \{(1,2),(1,3),(2,3)\}$

$R = \{\underbrace{(1,\overset{\notin A}{4})}_{\notin A\times A},(2,3)\}$ <u>not</u> a rel^n on A

$A = \{1,3,5\}$, $B = \{2,4,6\}$

$R = \{(1,4),(3,6),(5,2)\}$: rel from A to B



<u>Def</u> The <u>inverse</u> of a relation R:

$R^{-1} = \{(x,y) : (y,x)\in R\}$

$$(x,y) \in R^{-1} \longleftrightarrow (y,x) \in R$$

$$R = \{(1,1), (2,2), (3,3)\}$$

$$R^{-1} = \{(1,1), (2,2), (3,3)\} = R$$

$$R = \{(x,y) \in A \times A : x < y\}$$

$$= \{(1,2), (1,3), (2,3)\}$$

$$R^{-1} = \{(x,y) \in A \times A : y < x\}$$

$$= \{(2,1), (3,1), (3,2)\}$$

inverse $\neq$ negation

just flip x, y

Def (Properties of relations)

Let R be a relation on set A.

a) R is reflexive if xRx for all $x \in A$.

"$=$" on $\mathbb{Z}$ is reflexive:

$x = x$ for all $x \in \mathbb{Z}$

"$<$" on $\mathbb{Z}$ is not reflexive:

$1 \not< 1$, $1 \in \mathbb{Z}$

"$\leq$" on $\mathbb{Z}$ is reflexive:

$x \leq x$ for all $x \in \mathbb{Z}$

"$|$" on $\mathbb{Z}$ is reflexive:

divides

$x \mid x$ for all $x \in \mathbb{Z}$

$(x = x(1))$

b) R is symmetric if for all $x, y \in A$,

$\boxed{xRy \text{ implies } yRx}$

"$=$" on $\mathbb{Z}$ is symmetric:

if $x = y$, then $y = x$

"$<$" on $\mathbb{Z}$ is not symmetric:

$1 < 2$ but $2 \nless 1$

"$\leq$" on $\mathbb{Z}$ is not symmetric:

$1 \leq 2$ but $2 \nleq 1$

"$|$" on $\mathbb{Z}$ is not symmetric:

$1 \mid 2$ but $2 \nmid 1$

c) R is transitive if for all $x, y, z \in A$,

xRy and yRz imply xRz

"$=$" on $\mathbb{Z}$ is transitive:

If $x=y$ and $y=z$, then

$x = y = z$, and $x=z$

"$<$" on $\mathbb{Z}$ is transitive:

If $x<y$ and $y<z$, then

$x<y<z$ and $x<z$.

"$\leq$" on $\mathbb{Z}$ is transitive:

If $x \leq y$ and $y \leq z$,

then $x \leq y \leq z$, $x \leq z$

"$|$" on $\mathbb{Z}$ is transitive:

"divides" (proved in lessons)

If $a|b$ and $b|c$, then $a|c$

$b = an$ $\qquad c = bm$

$c = bm = a(nm)$